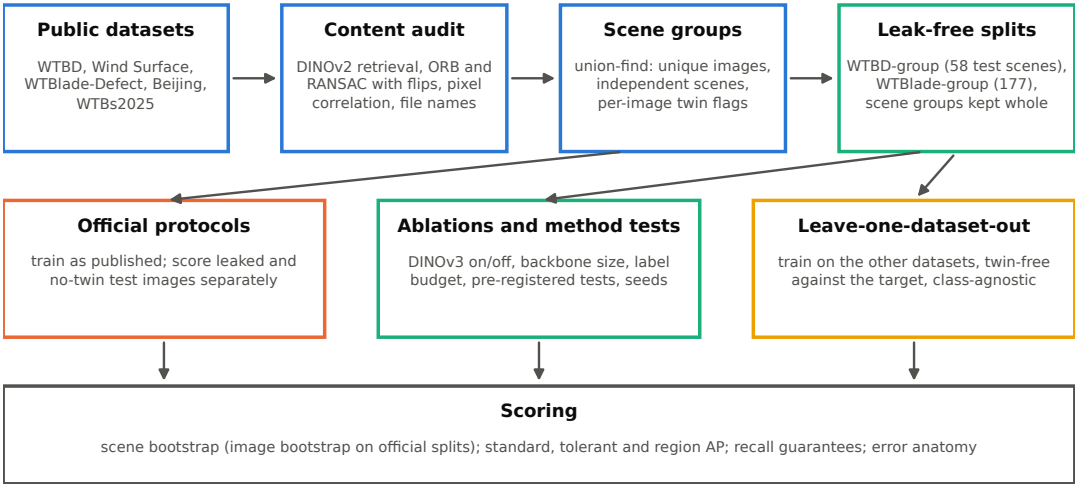
Public datasets
WTBD, Wind Surface, WTBlade-Defect, Beijing, WTBs2025
Content audit
DINOv2 retrieval, ORB and RANSAC with flips, pixel correlation, file names
Scene groups
union-find: unique images, independent scenes, per-image twin flags
Leak-free splits
WTBD-group (58 test scenes), WTBlade-group (177), scene groups kept whole
Official protocols
train as published; score leaked and no-twin test images separately
Ablations and method tests
DINOv3 on/off, backbone size, label budget, pre-registered tests, seeds
Leave-one-dataset-out
train on the other datasets, twin-free against the target, class-agnostic
Scoring
scene bootstrap (image bootstrap on official splits); standard, tolerant and region AP; recall guarantees; error anatomy